White paper: the three carryover analysis specifications, and why these three

2026-08-05 10:02 PDT; restructured 2026-08-05 10:46 PDT

pmsimstats team. A design-justification note for the carryover-sensitivity manuscript (`analysis/report/02-carryover-sensitivity/report.Rmd`). It names the three analysis-model specifications the study compares, states them in the canonical notation of `analysis/report/NOTATION.md`, and sets out why these three are the right set to evaluate. It also records why a fourth candidate is retained as a reported side-check rather than an arm, identifies the one cell the set still leaves empty, and lists four additions the comparison requires of the canonical notation.

Revision note. This document originally presented the middle arm as the Jones and Kenward lagged-treatment indicator. That arm has been replaced by the linear washout term supplied by the reference implementation, for the reasons in Section 5. The lagged indicator is retained as a side-check (Section 3.4).

Intended reader. A statistically literate physician: someone comfortable with regression and with the idea of an interaction term, but who should not be expected to supply the simulation-methods background. Statistical terms are glossed at first use.

1. The clinical problem, and the choice it forces

In an aggregated N-of-1 trial, each patient is crossed repeatedly between active drug and comparator, and many such single-patient series are pooled into one mixed model. The question these trials exist to answer is whether a baseline measurement predicts who benefits, that is, whether a **predictive biomarker** identifies the patients in whom the drug works. Statistically this is a biomarker-by-treatment **interaction**: the treatment effect is allowed to differ according to the patient's baseline biomarker value.

The complication is that drugs do not stop working the moment they are stopped. The effect decays over days to weeks, so measurements taken during a nominally off-drug period still reflect partial drug exposure. This residual effect is called **carryover**. In the prazosin and post-traumatic stress disorder setting this program is calibrated to, the relevant half-life is on the order of half a week to a week, while measurements are weekly, so the first off-drug measurement typically still carries roughly half the drug effect.

This forces a decision on the analyst that has no default answer. The model needs a variable representing drug exposure at each occasion, and coding it as simply on or off is one choice among several. The manuscript exists to determine which choice performs best; this note explains why the field was narrowed to three.

1.1 The three trial designs

The comparison runs under three protocols, which differ in how the on-drug and off-drug occasions are arranged. Because the whole problem concerns what happens *after* the drug stops, a design generating more post-discontinuation observations gives the analyst more carryover to handle, and the ranking of the specifications depends on which design is used. All three use eight measurement occasions per patient. Patients are randomized between two or four **paths**, meaning fixed schedules of on-drug and off-drug periods.

- **CO (crossover).** The traditional two-period within-patient crossover. One path takes drug for the first four visits and stops; the other takes comparator first and starts drug at visit five. Half the patients therefore never discontinue at all.
- **OL+BDC (open-label plus blinded discontinuation).** Open titration on active drug, then a blinded switch to placebo. Both paths discontinue, but late, so only two or three visits follow.
- **Hybrid.** The Hendrickson design and this program's reference: open-label titration, blinded discontinuation, then a brief crossover. Four paths, and the richest in on-off transitions.

2. Notation

Symbols follow `analysis/report/NOTATION.md`, the compendium's single source of truth. The plain-language column is added here and is not part of that file.

Symbol	`NOTATION.md` meaning	In plain terms
Y_{it}	symptom score for patient i at timepoint t (code `Sx`)	the outcome measured
B_i	pre-treatment biomarker (code `bm`)	the baseline test result
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$	the same, in standard-deviation units
D_{it}	binary drug state: 1 on drug, 0 off	the simple on/off switch
$D_{bc,it}$	exposure-decayed indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off (code `Dbc`)	a dimmer fading after the drug stops
t_{sd}	time since discontinuation (code `tsd`)	weeks since the drug was stopped
$t_{1/2}$	carryover half-life (code `carryover_t1half`)	time for residual effect to halve
λ	decay rate, $\ln 2/t_{1/2}$	how fast the residual effect fades
u_i	patient random intercept	each patient's own baseline level
ε_{it}	observation-level residual	unexplained variation
c_{bm}	moderation parameter, covariance-moderation architecture	how strongly the biomarker predicts benefit
σ_{BR}	standard deviation of the response component	the scale the effect is expressed on
θ_{true}	calibrated true value, $-c_{bm}\sigma_{BR}$	the correct answer to check against
N	total patients, across all paths	trial size

A hat marks an analyst-side quantity, so $\hat{t}_{1/2}$ is the half-life the analyst assumes while $t_{1/2}$ is the one the data were generated with. Per rule 5 of `NOTATION.md`, response components are non-negative *reductions* in severity, so a beneficial treatment effect lowers Y_{it} and interaction coefficients are negative.

Four symbols used here are not in `NOTATION.md`; Section 9 proposes them formally. They are L_{it}, the lagged just-off-drug indicator; $\beta_{bm:D}$ disambiguated from $\beta_{bm:D_{bc}}$ according to which exposure variable carries the interaction; the main-effect coefficients β_b, β_t, β_D, β_L, β_{sd}; and the stored specification code `E4`. Note that β_b, the biomarker main effect, is a different object from β_{bm}, which `NOTATION.md` reserves for the mean-moderation architecture's moderation parameter.

Abbreviations, per the `NOTATION.md` glossary: **CO** (crossover), **BDC** (blinded discontinuation), **OL+BDC**, **Hybrid**, **LME** (linear mixed-effects model), **MCSE** (Monte Carlo standard error), **ADEMP** (the Morris, White and Crowther reporting framework).

All three specifications share one model,

$$Y_{it} = \beta_0 + \beta_b\, b_i + \beta_t\, t + \beta_D\, X_{it} + \beta_{bm:X}\, b_i X_{it} + u_i + \varepsilon_{it},$$

fitted as an LME with a patient random intercept and `corCAR1` residual correlation, which lets measurements closer in time be more similar. In words: the outcome depends on the biomarker, on time, on drug exposure, and on the product of biomarker and exposure. That product term is the **estimand**, the specific quantity the trial is trying to estimate; it says how much the treatment effect changes per standard deviation of the biomarker. The specifications differ only in how X_{it} is built and in whether a nuisance term is added.

3. The three specifications

3.1 Unadjusted (stored code `E1`)

$$X_{it} = D_{it}, \qquad D_{it} \in \{0, 1\}$$

In code, `Sx ~ bm + t + Db + bm:Db`. Carryover is not represented. A measurement one week after stopping, still carrying about half the drug effect, is coded exactly like one taken after full clearance. The estimand is $\beta_{bm:D}$, the coefficient on $b_i D_{it}$.

This is what an analyst gets by default from a standard mixed model without thinking about carryover, which is why it belongs in the comparison: it is the benchmark any remedy must beat.

3.2 Washout-adjusted (stored code `E4`)

$$X_{it} = D_{it}, \qquad \text{plus } \beta_{sd}\, t_{sd,it} \text{ as a nuisance main effect}$$

In code, `Sx ~ bm + t + Db + bm:Db + tsd`. Exposure stays binary, but the model is given the elapsed time since discontinuation as a covariate, so it can estimate how the average outcome drifts across the washout without being told how fast the drug clears. This is the alternative the reference implementation provides (`implementations/original/R/lme_analysis.R`, `simplecarryover = TRUE`).

Two properties matter. It consumes no half-life, so it is completely unaffected by what the analyst assumes about the pharmacokinetics. And because t_{sd} enters the average-outcome part of the model only, with no $b_i t_{sd}$ product, it can absorb mean-level carryover but cannot repair distortion of the *biomarker* effect. The estimand is $\beta_{bm:D}$, the same quantity Unadjusted targets.

3.3 Exposure-weighted (stored code `E2`)

$$X_{it} = D_{bc,it} = \begin{cases} 1 & \text{if } D_{it} = 1, \\ \exp(-\hat{\lambda}\, t_{sd,it}) & \text{if } D_{it} = 0, \end{cases} \qquad \hat{\lambda} = \frac{\ln 2}{\hat{t}_{1/2}}$$

In code, `Sx ~ bm + t + Dbc + bm:Dbc`. Instead of an on/off switch, exposure is a dimmer starting at 1 and halving every $\hat{t}_{1/2}$ weeks. A measurement one week after stopping, at an assumed half-life of one week, enters as 0.5 rather than 0. This is the current pmsimstats-ng default.

It is the only one of the three requiring a commitment to both a decay shape and a half-life. Because those enter through the variable's values rather than the formula, it is effectively a different model for every assumed half-life though written identically each time. Its estimand is $\beta_{bm:D_{bc}}$, the coefficient on $b_i D_{bc,it}$, which is *not* the same quantity as $\beta_{bm:D}$.

3.4 Retained as a side-check: Lag-adjusted (stored code `E3`)

$$L_{it} = D_{i,t-1}\,(1 - D_{it})$$

In code, `Sx ~ bm + t + Db + bm:Db + L`. The indicator equals 1 at the first off-drug measurement following an on-drug one, and 0 elsewhere. This is the classical crossover remedy of Jones and Kenward, and it occupies the same conceptual slot as Washout-adjusted: a nuisance main effect, no half-life, estimand $\beta_{bm:D}$. It differs in being indexed by position in the visit sequence rather than by elapsed time, so it marks one occasion and is constant thereafter.

It is not an arm of the study, for the reasons in Section 5.1, but its results are already present in every stored summary and are reported as a side-check. Section 6 gives them.

3.5 The three codings on one patient

The differences are clearest on a single schedule. Both tables assume a half-life of one week. Verified by direct computation from the design definitions in `simulation-core.R`.

OL+BDC, path A. Drug stopped after week 18.

Week	On drug	t_{sd}	Unadjusted D_{it}	Washout t_{sd}	Lag flag L_{it}	Exposure-weighted $D_{bc,it}$
4 to 18	yes	0	1	0	0	1
19	no	1	0	1	1	0.50
20	no	2	0	2	0	0.25

At week 19 the patient still has about half the drug effect on board. Unadjusted records a 0, exactly as for a drug-free patient. Washout-adjusted also records 0 for exposure, but hands the model the elapsed time so the average outcome may drift. Only Exposure-weighted records a value reflecting the residual exposure, 0.50.

Hybrid, path A. Stopped after week 10, restarted at 16, stopped after 16.

Week	On drug	t_{sd}	Unadjusted D_{it}	Washout t_{sd}	Lag flag L_{it}	Exposure-weighted $D_{bc,it}$
4 to 10	yes	0	1	0	0	1
11	no	1	0	1	1	0.50
12	no	2	0	2	0	0.25
16	yes	0	1	0	0	1
20	no	4	0	4	1	0.0625

This path shows why the lag flag was replaced. Weeks 11 and 20 both receive $L_{it} = 1$, each being the first visit after a discontinuation, so the model gives them one shared coefficient. Yet week 11 retains about 50% of the drug effect and week 20 about 6%, an eightfold difference. Washout-adjusted distinguishes them, 1 against 4, and Exposure-weighted distinguishes them further, 0.50 against 0.0625. The lag flag alone cannot tell them apart.

4. Why these three

4.1 They span the three positions on washout

Write $\phi(t_{sd})$ for the fraction of drug effect still present t_{sd} weeks after stopping. Every analyst takes a position on ϕ, and only three are qualitatively distinct:

- **Deny it.** Assume $\phi \equiv 0$; exposure is strictly on or off. This is Unadjusted.
- **Estimate it.** Leave the shape unspecified but let the data say how the outcome drifts across the washout. This is Washout-adjusted.
- **Assume it.** Commit to a decay curve and rate, $\phi = \exp(-\hat{\lambda} t_{sd})$. This is Exposure-weighted.

The three are therefore one representative of each available stance toward the unknown washout, ordered by how much is assumed.

A formal relationship makes the ladder continuous. As the assumed half-life goes to zero the dimmer collapses to the on/off switch, since $\hat{t}_{1/2} \to 0$ gives $\hat{\lambda} \to \infty$ and hence $D_{bc,it} \to D_{it}$. **Unadjusted is therefore the limiting case of Exposure-weighted at an assumed half-life of zero**, not a separate method outside it. As $\hat{t}_{1/2} \to \infty$ the dimmer never dims, the on-off contrast vanishes, and nothing is estimable. The manuscript's half-life-mis-specification sweep is thus a traverse along a path anchored at Unadjusted.

4.2 They separate the two ways carryover does damage

This is the strongest reason for the particular set, because it makes the comparison a controlled experiment rather than a horse race.

Carryover degrades the interaction test through two mechanisms:

- **Dilution.** The exposure variable is a mis-measured version of what the patient received. Contaminated occasions are entered as drug-free, so the estimated interaction is pulled toward zero, the same phenomenon clinicians know as regression dilution, where a single noisy blood-pressure reading understates a true association. The signal shrinks.
- **Added noise.** Those occasions also differ from one another in ways the model does not describe, some being one week off drug and others four. That variation is absorbed into ε_{it}, inflating the standard error of the interaction estimate. The noise grows.

Power falls on both counts at once. The three arms occupy three cells:

Specification	Dilution	Added noise
Unadjusted	not addressed	not addressed
Washout-adjusted	not addressed	addressed, graded across the whole washout
Exposure-weighted	addressed	addressed
(Lag-adjusted, side-check)	*not addressed*	*addressed at one occasion only*

The placements follow from where each term sits. Washout-adjusted adds t_{sd} to the average-outcome part only, so it absorbs noise while the interaction stays on D_{it} and dilution is untouched. Exposure-weighted replaces the variable *inside* the interaction, so it addresses dilution directly and mops up the noise as a by-product.

Two consequences follow:

1. **Comparing Washout-adjusted with Unadjusted isolates the noise mechanism.** The two estimate the same quantity, are fitted to the same simulated patients, and differ by exactly one term. Any difference is attributable to the noise mechanism alone. Controlled contrasts this clean are rare in a simulation study.
2. **What is left over is interpretable.** Whatever advantage Exposure-weighted shows beyond that is the value of fixing dilution. The design reveals not only which method wins but which mechanism matters, which is the more portable finding.

4.3 Each is a working default, and the set is now complete

Unadjusted is ordinary applied practice and the default of any off-the-shelf mixed model. Washout-adjusted and Exposure-weighted are the two options the reference implementation actually provides (Section 5.2). The set therefore evaluates that implementation completely, rather than two-thirds of it plus a substitute.

This involves a deliberate trade, which should be stated plainly. The crossover tradition's own remedy, the lagged indicator, moves from a headline arm to a side-check. What is bought is that the study now tests every option the original authors offered; what is given up is that the crossover literature is represented by a reported result rather than by a full arm. Section 6 reports that result, so the tradition is answered, but with less prominence.

4.4 They are equally estimable

All three fit in all three designs at both sample sizes, need no data beyond what the trial collects, and add at most one term. Verified for Unadjusted and Exposure-weighted by recomputation from the stored summaries: across the full 540-cell grid neither produced a missing power value or estimate, and the same holds for the retained side-check. Washout-adjusted requires at least one post-discontinuation occasion, which every path of every design except CO path B provides; that path never discontinues, so it contributes to the other terms but not to β_{sd}. This has not yet been verified by a run.

5. Why the middle arm was changed, and what is still missing

5.1 Why Lag-adjusted was replaced

Three reasons, in increasing order of weight.

Provenance. The lagged indicator is not inherited from the reference implementation. It was introduced by this project in `simulation-core.R` following Jones and Kenward. Reporting it as the comparator while leaving the original authors' own alternative untested is hard to defend.

Resolution. As Section 3.5 shows, the flag marks one occasion and cannot distinguish a visit one week after stopping from one four weeks after. In the Hybrid design it assigns a single coefficient to occasions differing eightfold in residual exposure.

It may have produced a false negative. Recomputed from the stored summaries, Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 cells in scope (Section 6). The manuscript currently reads that as evidence that the noise mechanism is negligible, and therefore that all of Exposure-weighted's advantage comes from fixing dilution. That inference is only as good as the arm it rests on. The `quick-sim` outputs show the reference implementation's t_{sd} arm reaching an *identical point estimate* to its unadjusted arm while achieving materially higher power, which is the exact signature of a gain through the noise channel with dilution untouched. If that reproduces here, the noise mechanism is not negligible; a binary flag was simply too crude to exploit it, and the manuscript's mechanistic conclusion would change.

5.2 The reference implementation's three options

`implementations/original/R/lme_analysis.R` supports three carryover representations, assembled at lines 142 to 153 with the exposure variable built at 108 to 118. None is a lagged indicator.

Setting	Fitted exposure terms	This study
`carryover_t1half = 0`, `simplecarryover = FALSE`	$D_{it} + b_i D_{it}$	Unadjusted
`carryover_t1half = 0`, `simplecarryover = TRUE`	$D_{it} + b_i D_{it} + \beta_{sd} t_{sd,it}$	Washout-adjusted
`carryover_t1half > 0`	$D_{bc,it} + b_i D_{bc,it}$	Exposure-weighted

The first and third are mutually exclusive with the second by construction: supplying both a non-zero half-life and `simplecarryover = TRUE` raises an error (lines 67 to 68).

A divergence to disclose. Hendrickson's default `useDE = TRUE` appends the expectancy covariate De wherever expectancy varies, which it does in OL+BDC and Hybrid though not CO. The specifications here omit it: `simulation-core.R` passes `expectancies` to the design builder, so expectancy shapes the simulated data, but De never enters any fitted formula. It applies equally to all arms so should not disturb their ranking, but the fitted model is not Hendrickson's, and the agreement between this study's 0.860 and Hendrickson's reported 0.82 has not been checked against that difference.

5.3 The cell that remains empty

Replacing the middle arm improves it but does not fill the gap in Section 4.2. Washout-adjusted still carries the interaction on D_{it}, with t_{sd} entering as a main effect only, so dilution remains unaddressed by any assumption-free method. The missing specification is now

$$Y_{it} = ... + \beta_D D_{it} + \beta_{sd} t_{sd,it} + \beta_{bm:D}\, b_i D_{it} + \beta_{bm:sd}\, b_i t_{sd,it} + u_i + \varepsilon_{it},$$

in code `Sx ~ bm + t + Db + tsd + bm:Db + bm:tsd`. This is a more interesting candidate than the $b_i L_{it}$ term it replaces, because it is essentially a linear approximation to what Exposure-weighted does parametrically: it lets the biomarker effect decline across the washout without specifying a rate. Adding it would complete the ladder of Section 4.1.

It is not proposed for this manuscript. If the grid stays at three, the Methods section should state that every assumption-free specification evaluated here enters carryover as a main effect only, and that the study therefore bounds what such methods can achieve against dilution rather than settling it.

5.4 Candidates set aside

- **Washout deletion**, discarding off-drug measurements within a window of stopping. Exposure weighting with crude 0/1 weights and an arbitrary cutoff; discards observations the others use.
- **Estimating the half-life from the data.** The principled answer to Exposure-weighted's main weakness and the natural next step, but the half-life is poorly determined at $N = 35$ from a handful of off-drug measurements, introducing a convergence-failure mode the others do not have.
- **A full set of lag indicators**, one per off-drug occasion. The window is three to four measurements long, so a full set is nearly indistinguishable from the time term t.

6. What is known, and what must still be checked

Epistemic status: the Lag-adjusted and Unadjusted figures below are recomputed directly from the stored cell-level summaries under `analysis/data/`. **No Washout-adjusted results exist yet**; the specification has been added to `fit_spec()` but not run.

How the comparison is scored. Following ADEMP: **power**, how often a real effect is detected, target 0.80; **Type I error**, how often an effect is declared when none exists, which should sit at 0.05; **bias**, the average gap between estimate and truth; **coverage**, how often the 95% interval contains the truth, target 0.95; **MSE**, combining bias and spread; and **MCSE**, the simulation's own margin of error, below which differences should not be interpreted. A **cell** is one combination of design, sample size, biomarker strength, decay shape and half-life, run 500 times.

The side-check result. Within the 216-cell scope, Lag-adjusted and Unadjusted are statistically indistinguishable. Mean power differs by 0.003 against an MCSE of about 0.018, only 3 of 216 conditions differ by more than 0.02, and mean bias (0.373 versus 0.372), MSE (1.885 versus 1.876) and coverage (0.936 versus 0.938) agree to three decimal places. The same equality holds in every sensitivity block and in the cluster-robust cells. The classical lagged device therefore provides no measurable benefit over ignoring carryover altogether, which is a reportable finding in its own right.

What the reference implementation's output suggests. Its t_{sd} arm leads its unadjusted arm by 1.3 to 6.5 percentage points on paired tests, with an identical point estimate (-1.61 in both) and nearly identical empirical spread. Identical estimate with higher power means the gain came through the noise channel, consistent with the classification in Section 4.2.

Two checks that must pass before that is believed. First, the `quick-sim` run uses a different parameterization and a different code path, so it establishes only that the term can work, not that it will here. Second, and more seriously, its t_{sd} arm carried the highest false-positive rate of its three, up to 0.053 against 0.037 where 0.05 is nominal. If Washout-adjusted proves anticonservative under this study's parameterization, part or all of its power advantage is not real. Type I error must therefore be reported alongside power, and no power gain should be claimed until that check passes.

7. Which comparisons this design supports

Power and Type I error are comparable across all three. These are properties of the test, not of the coefficient tested, so they can be compared freely even though the three do not all estimate the same quantity.

Bias, MSE and coverage are comparable only within the Unadjusted and Washout-adjusted pair. Those two estimate the coefficient on $b_i D_{it}$; Exposure-weighted estimates the coefficient on $b_i D_{bc,it}$. All are scored against the single target $\theta_{\text{true}} = -3.6$, correct for Exposure-weighted but not for the others. Exposure-weighted will therefore show near-zero bias and nominal coverage while the others show substantial bias and under-coverage, and that gap is largely a statement about which quantity each targets. Those columns must not be used to rank the three.

The Washout-adjusted versus Unadjusted contrast is the controlled one. Same estimand, same simulated patients, one term of difference. Whatever separates them is the noise mechanism, cleanly identified. This is the most trustworthy comparison in the study.

Conclusions about assumption-free methods are bounded by the main-effect form. Neither Washout-adjusted nor the retained side-check includes a biomarker interaction with the carryover term, so a null result for either says that a carryover *main effect* does not recover the diluted signal. It does not establish that no assumption-free method can (Section 5.3).

8. Summary

The three specifications are Unadjusted, which codes exposure as an on/off switch; Washout-adjusted, which adds elapsed time since stopping as a covariate; and Exposure-weighted, which replaces the switch with a dimmer fading at an assumed half-life. They are the right three because they occupy the three distinct positions on the unknown washout curve, because two share an estimand and so isolate one damage mechanism under otherwise identical conditions while the third addresses the other, and because the set now covers every option the reference implementation provides.

The middle arm was changed from the Jones and Kenward lagged indicator because that indicator is a project invention rather than the published alternative, because it cannot distinguish occasions differing eightfold in residual exposure, and because its null result may have been an artifact of that crudeness rather than evidence about the mechanism. It is retained as a reported side-check. The set still leaves one cell empty, at $b_i t_{sd}$, which should be disclaimed in the Methods if it is not filled.

9. Proposed additions to NOTATION.md

9.1 Add L_{it} and t_{sd} as model terms. The lagged just-off-drug indicator has no symbol in Part 1. Proposed: $L_{it} = D_{i,t-1}(1 - D_{it})$, equal to 1 at the first off-drug occasion following an on-drug one (code column `L`). t_{sd} is already listed as a data-generating quantity; the entry should note that it also serves as an analysis-model covariate.

9.2 Split the estimand symbol. Part 1 carries a single entry, $\beta_{bm:D}$, glossed as the interaction coefficient and annotated `bm:Dbc`. That conflates two coefficients. Proposed: $\beta_{bm:D}$ for the coefficient on $b_i D_{it}$ (`bm:Db`), the estimand of Unadjusted, Washout-adjusted and the side-check; and $\beta_{bm:D_{bc}}$ for the coefficient on $b_i D_{bc,it}$ (`bm:Dbc`), the estimand of Exposure-weighted. This distinction is what licenses the controlled comparison of Section 4.2 and why the bias and coverage columns are comparable within one pair but not across to the third.

9.3 Add the main-effect coefficients. Part 1 has no symbols for the intercept or non-interaction terms. Proposed: β_0 intercept, β_b biomarker, β_t time, β_D drug exposure, β_L lagged indicator, β_{sd} linear time since discontinuation. The β_b entry should note that it is distinct from β_{bm}, reserved for the mean-moderation architecture's moderation parameter.

9.4 Register the `E4` stored value. The Part 2 table lists `spec` as taking `E1`, `E2`, `E3`. Washout-adjusted adds `E4`. It is a new code rather than a redefinition of `E3` deliberately: redefining would silently change the meaning of `E3` in the twenty-one migrated `.rds` files and in whatever papers 01 and 10 quote, which is the precise failure mode Part 2 exists to prevent. The Part 3 glossary entry for analysis specifications needs the fourth line accordingly.

9.5 A policy tension to resolve before renaming. Part 2 requires that a stored value doubling as a reporting label be kept identical to it, and records `spec` as unmapped. That rule was enforced on 2026-07-31, when twenty-one `.rds` files were migrated from `A1/A2/A3` to retire a display-time mapping, so renaming reintroduces the layer that migration removed. The same file supplies the governing precedent, however: the architectures were renamed to mechanism names while their stored values were left untouched and the mapping recorded. Follow that precedent rather than migrating twice. Keep `E1` to `E4` stored permanently, apply display names through one shared helper, and mark the Part 2 row as mapped.

Rendered on 2026-08-05 at 10:46 PDT. Source: `~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/02-carryover-sensitivity/whitepaper-three-analysis-specifications.md`